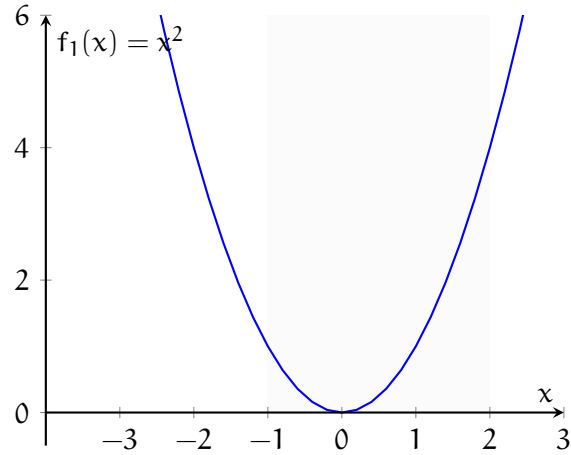


We first plot the function f_1 to have an intuition.



The infimum of x^2 on $\mathbb{R}$ is 0:

$$\inf_{x \in \mathbb{R}} x^2 = 0.$$

Indeed, for each $M > 0$, there exists

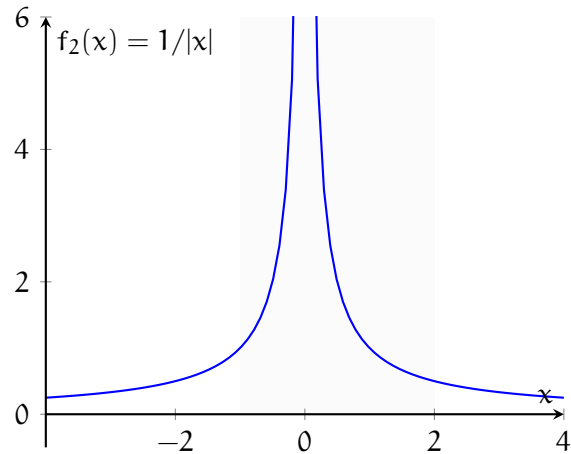
$$y = \frac{\sqrt{M}}{2}$$

such that $y^2 = \frac{M}{4} < M$. The function has also a minimum at $x^* = 0$, as

$$f_1(x^*) = \inf_{x \in \mathbb{R}} x^2 = 0.$$

When the constraints are introduced, the same arguments can be used to reach the same conclusions: the infimum is 0, and $x^* = 0$ is an optimum.

We now analyze the function f_2 .



The infimum of $1/|x|$ on $\mathbb{R}$ is 0:

$$\inf_{x \in \mathbb{R}} \frac{1}{|x|} = 0.$$

Indeed, for each $M > 0$, there exists

$$y = \frac{2}{M}$$

such that $f_2(y) = \frac{M}{2} < M$. However, there is no minimum, as there is no x such that

$$\frac{1}{|x|} = 0.$$

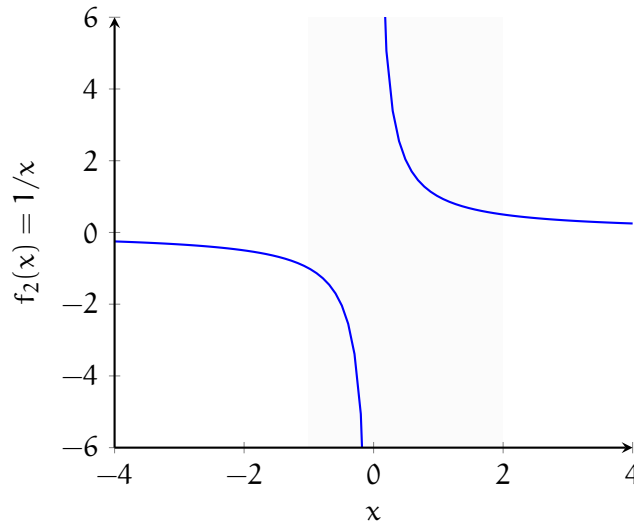
The infimum of $1/|x|$ on $Y = \{x \in \mathbb{R} \mid -1 \leq x \leq 2\}$ is 0.5. Indeed, for each $M > 0.5$, there exists

$$y = 2 \in Y$$

such that $f_2(y) = 0.5 < M$. And the minimum is $x^* = 2$, as

$$f_2(x^*) = 0.5 = \inf_{x \in Y} \frac{1}{|x|}.$$

Finally, we analyze the function f_3 .



There is no infimum on $\mathbb{R}$. Indeed, the function is not bounded from below. As the infimum is the best lower bound, and there is no lower bound, there is no infimum. Consequently, there is no optimum either. As the function is also not bounded from below on the interval $[-1 : 2]$, we reach the same conclusions for the constrained case.